

A Deep-Learning-Assisted Homogenization Framework for Permeability Upscaling in Porous Media

Introduction

Homogenization theory provides a rigorous framework to upscale pore-scale flow information to macroscopic effective parameters, such as permeability and diffusivity tensors (Auriault et al. 2010). In porous media, homogenization establishes a systematic link between microscale velocity/pressure fields and equivalent continuum-scale flow equations, offering a physically consistent and interpretable route for effective property estimation.

In practice, homogenization typically requires solving auxiliary (cell) problems on finely discretized representative elementary volumes (REVs) to obtain pore-scale velocity and pressure fields as well as the corresponding effective coefficients (Yan et al. 2019). Although direct numerical simulation (DNS) of flow in complex porous structures can provide high-fidelity microscale transport information, its computational cost is often prohibitive for large-scale or high-resolution applications (Liu et al. 2023). As a result, computing effective permeability through DNS-based averaging is impractical for real-time or multiscale scenarios. In recent years, neural-operator-based surrogate models for porous-media flow have therefore been actively explored, including physics-informed variants that enhance robustness and generalization under heterogeneous conditions (Yan et al. 2025).

To address this bottleneck, we develop an efficient deep-learning-assisted homogenization framework for permeability upscaling in complex porous media. Specifically, we employ a U-shaped Fourier Neural Operator (U-FNO) (Wen et al. 2022) to learn the mapping from pore geometry to the corresponding pore-scale steady-state velocity field, thereby replacing conventional DNS solvers. The predicted velocity field is then fed into the homogenization procedure to compute the effective permeability. This hybrid approach combines the speed and scalability of neural operators with the physical rigor of homogenization, enabling accurate and real-time permeability estimation across diverse microstructures. The proposed method provides a computationally efficient bridge between pore-scale simulations and continuum-scale models.

Method

We consider steady incompressible creeping flow in an ε -periodic porous medium, where ε denotes the ratio between the characteristic pore length and the macroscopic length. By applying classical two-scale homogenization to the pore-scale Stokes problem, the macroscopic flow is governed by Darcy's law:

$$\mathbf{u}(x) = -\frac{\mathbf{K}^{\text{hom}}}{\mu} \nabla p(x) \quad (1)$$

$$\nabla \cdot \mathbf{u}(x) = 0 \quad (2)$$

where $\mathbf{u}$ is the Darcy (superficial) velocity, p is the macroscopic pressure, μ is the dynamic viscosity, and $\mathbf{K}^{\text{hom}}$ is the homogenized permeability tensor. The homogenized permeability can be defined through a standard periodic cell problem posed on the representative unit cell Y , decomposed into a fluid region Y_f and a solid region Y_s with interface $\Gamma = \partial Y_f \cap \partial Y_s$. For a given macroscopic driving direction $\mathbf{e}_i$, the corresponding corrector velocity field $\mathbf{w}^{(i)}(y)$ and pressure $\pi^{(i)}(y)$, with $y = x/\varepsilon$, satisfy:

$$-\nabla_y \cdot \pi^{(i)} + \mu \nabla_y^2 \mathbf{w}^{(i)} = \mathbf{e}_i \text{ in } Y_f, \quad \nabla_y \cdot \mathbf{w}^{(i)} = 0 \text{ in } Y_f \quad (3)$$

$$\mathbf{w}^{(i)} = 0 \text{ on } \Gamma, \quad \mathbf{w}^{(i)}, \pi^{(i)} \in H_{\text{per}}^1(Y_f), \quad \langle \pi^{(i)} \rangle_{Y_f} = 0 \quad (4)$$

where $\mathbf{e}_i$ denotes the unit vector along the i -th coordinate direction and Y_f denotes the volume average over the fluid region Y_f . In this work, we focus on the effective permeability along the imposed flow direction $\mathbf{e}_i$. The homogenized permeability component in the i -th direction is then given by:

$$K_{ii}^{\text{hom}} = \frac{1}{|Y|} \int_{Y_f} w_i^{(i)}(y) dy \quad (5)$$

The overall framework proposed in this study is illustrated in Figure 1.

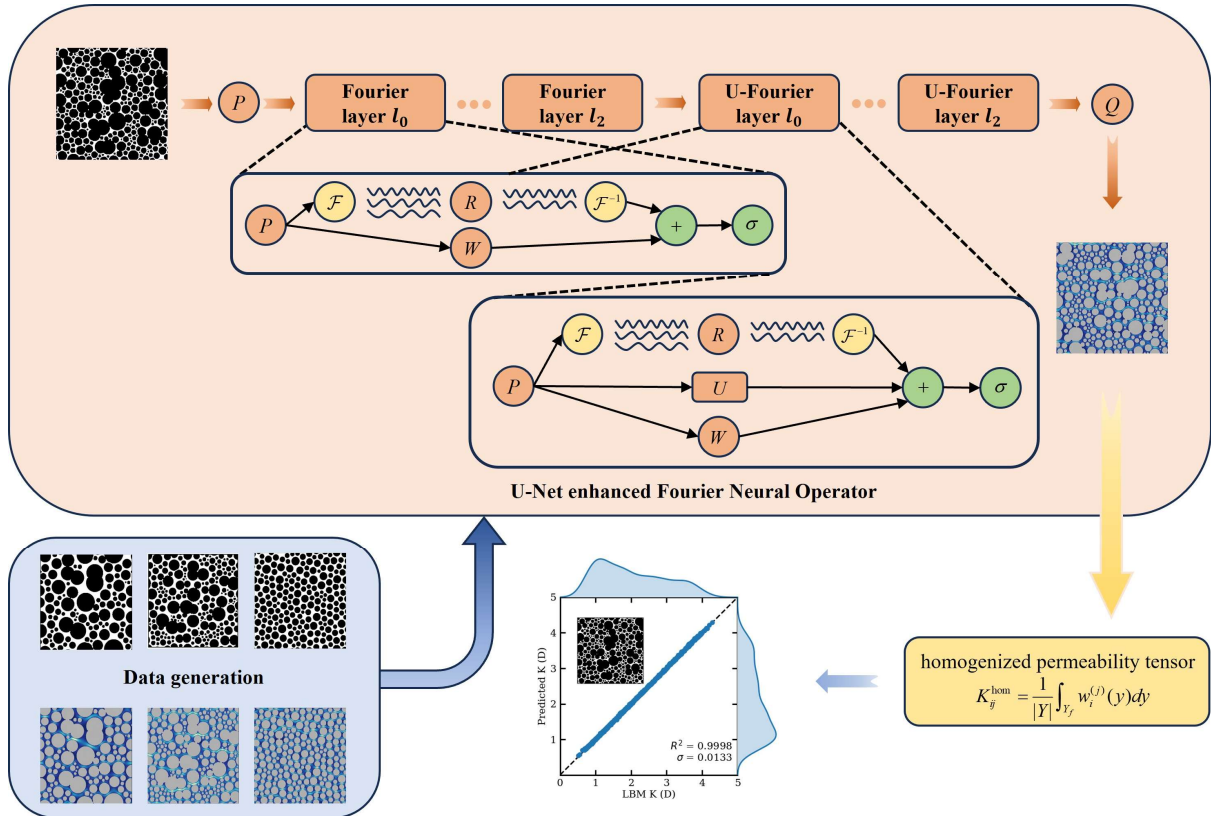


Figure 1 Workflow of a U-FNO-assisted homogenization framework for velocity-based permeability upscaling.

Examples

To evaluate the proposed framework, we consider steady single-phase flow through 2D disk-pack porous media and assess the accuracy of the predicted velocity fields and the resulting effective permeability. The dataset comprises six rock types (labeled 1–6) with prescribed structural variability, generated by varying both the number of disks and the variance of the lognormal distribution used to sample disk radii, thereby introducing different levels of structural heterogeneity. Specifically, the lognormal radius variance for rock types 1–6 is 0.05, 0.45, 0.15, 0.05, 0.35, and 0.35, and the corresponding disk counts are 100, 300, 100, 200, 100, and 200, respectively. Here, the variance parameter refers to the variance of the underlying lognormal radius distribution and controls the degree of particle-size polydispersity. Four rock types (3–6) are used for model development, with an 8:1:1 split for training, validation, testing, yielding (6000×4) samples in total. Two additional held-out rock types (1 and 2) are reserved for final evaluation, each containing 700 samples.

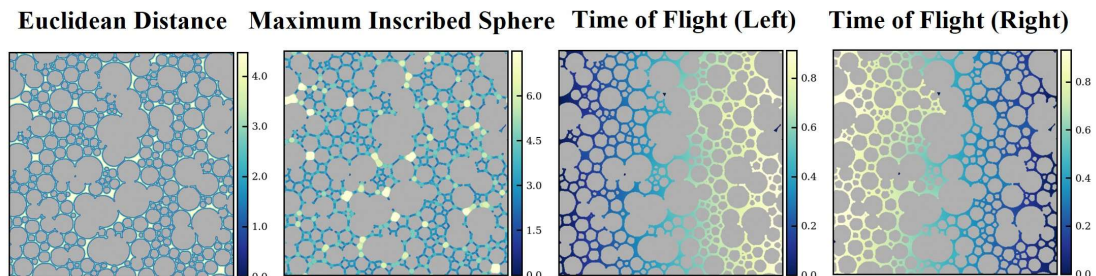


Figure 2 Input feature maps derived from the pore-solid geometry, including Euclidean Distance, Maximum Inscribed Sphere, and Time of Flight (Left/Right).

For each sample, the input comprises five channels at a resolution of (512×512): Euclidean Distance, Maximum Inscribed Sphere, Time of Flight (Left), Time of Flight (Right), and a binary pore–solid indicator. The Euclidean Distance, Maximum Inscribed Sphere, and Time of Flight (Left/Right)

features are shown in Figure 2. The U-FNO model retains 16 Fourier modes in each of the two spatial dimensions ($\text{modes1} = 16, \text{modes2} = 16$), uses a network width of 32, takes 5 input channels, and outputs the two velocity components: the x -direction velocity u and the y -direction velocity v . To balance variations in velocity magnitude across samples, the loss is computed on a per-sample basis and normalized using the sample-specific 99th-percentile velocities; that is, the prediction error is divided by these reference values.

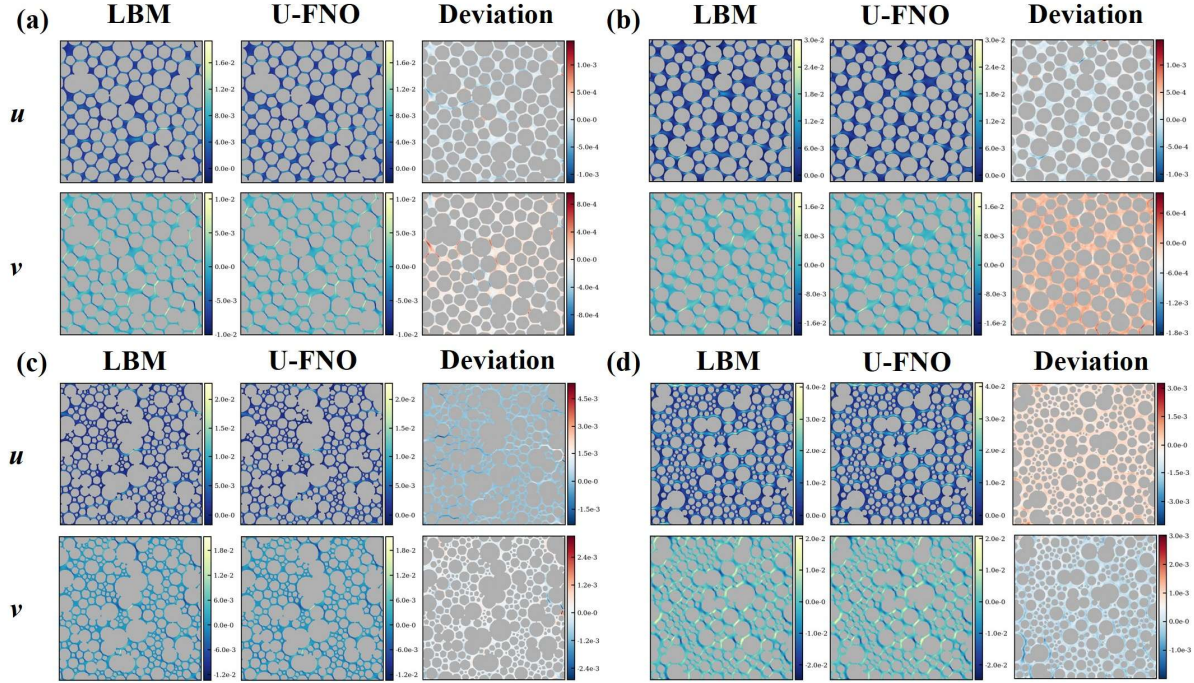


Figure 3 Comparison of velocity fields from LBM, U-FNO predictions, and corresponding error maps for held-out rock types.

Representative flow-field comparisons for the two held-out test rock types are shown in Figure 3. In Figure 3, panels (a) and (c) show low-porosity samples of rock types 1 and 2, and panels (b) and (d) show high-porosity samples of rock types 1 and 2. For each sample, we plot the two velocity components obtained from LBM simulations, the two velocity components predicted by the U-FNO surrogate, and the corresponding error maps for both velocity components.

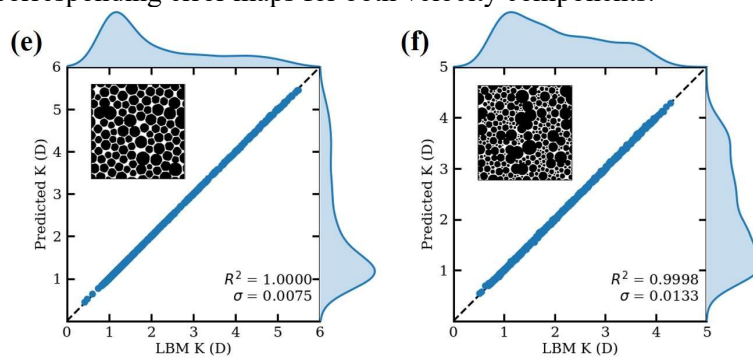


Figure 4 Parity plots of effective permeability predicted by the U-FNO-based homogenization and LBM reference for held-out rock types.

In addition, the permeability predicted via homogenization using the U-FNO velocity field is compared against the reference permeability computed from LBM. The parity plots for rock types 1 and 2 (Figure 4; panel (e) for rock type 1 and panel (f) for rock type 2) show excellent agreement, with $R^2 = 1.0000$ and $\sigma = 0.0075$ D for rock type 1, and $R^2 = 0.9998$ and $\sigma = 0.0133$ D for rock type 2. Here, permeability is reported in Darcy (D), where 1 D = 1000 mD. The overlaid marginal distributions along the top and right axes further illustrate that the predicted permeability closely matches the reference distribution across the full permeability range. These results indicate that the proposed pipeline can generalize well

to held-out microstructures while maintaining high accuracy in effective permeability estimation. From a computational perspective, our framework processes 1400 samples of held-out rock types in 435.92 s total (512×512 grid resolution, batch size = 8, single NVIDIA L20 GPU). GPU-accelerated XLB-LBM (Ataei and Salehipour 2024) takes 12572.32 s total for the same samples, corresponding to a speedup of $\sim 29\times$.

Conclusions

In this study, a deep-learning-assisted homogenization framework is developed for efficient permeability upscaling in complex porous media. The proposed approach replaces the conventional pore-scale DNS solver with a neural-network-based velocity surrogate, while retaining homogenization theory as a physically rigorous upscaling operator. Specifically, a U-FNO model is trained to predict steady-state pore-scale velocity fields from pore geometry, and the predicted velocity fields are subsequently used as inputs to the homogenization procedure to compute effective permeability. The performance of the framework is evaluated using two-dimensional disk-pack porous media with varying structural heterogeneity. Numerical results demonstrate that the predicted velocity fields closely match LBM reference solutions for held-out rock types, and that the effective permeability obtained through homogenization shows excellent agreement with LBM-based permeability, both in parity plots and marginal distributions. Moreover, In terms of efficiency, the proposed pipeline achieves an overall speedup of $\sim 29\times$ over GPU-accelerated XLB-LBM on 512×512 grids on a single NVIDIA L20 GPU. These results indicate that the proposed framework provides an efficient and physically consistent bridge between pore-scale simulations and macroscopic permeability estimation, and offers a promising pathway for scalable and interpretable multiscale flow modeling.

Acknowledgements

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